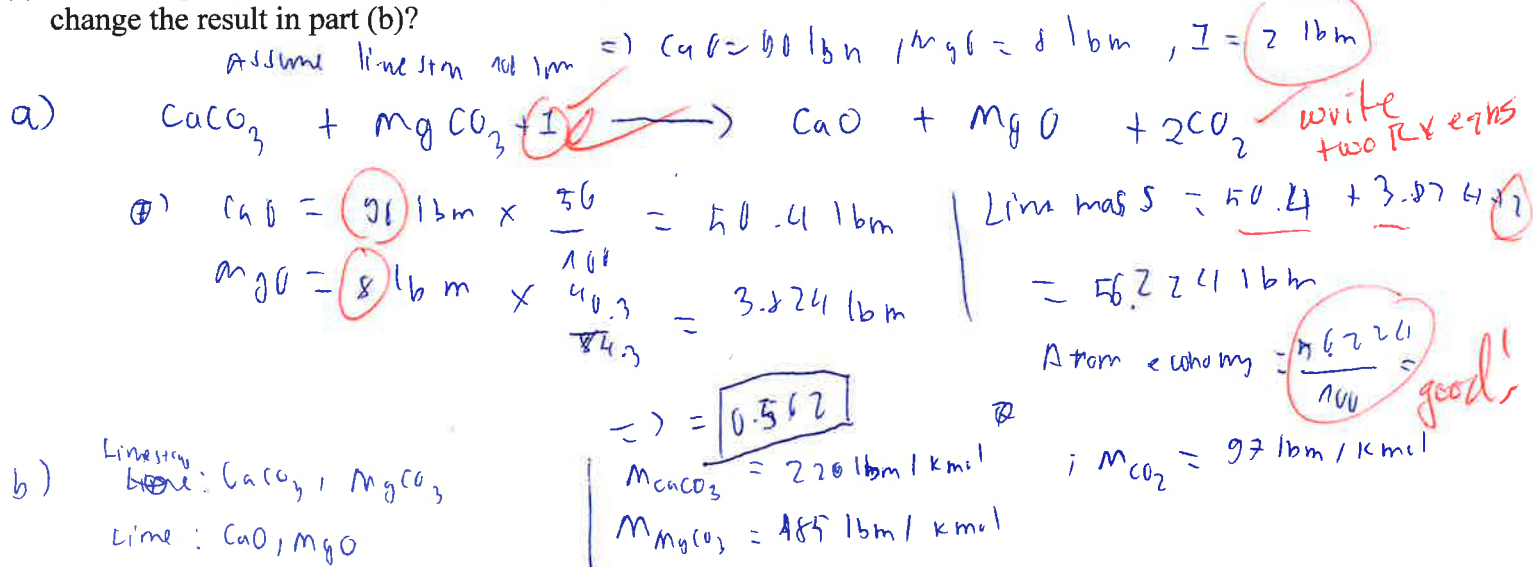


Limestone is available with the composition 90 wt% calcium carbonate (CaCO_3), 8 wt% magnesium carbonate (MgCO_3), and 2 wt% Insoluble Inerts (I). The limestone is heated to 1000°C in a continuous rotary lime kiln to drive off carbon dioxide (CO_2) and make lime. Lime will include the CaO and MgO produced plus all the impurities present in the limestone which remain after all the CO_2 has been driven off. Assume that both the calcium carbonate and the magnesium carbonate react completely to produce carbon dioxide.

Data: $M_{\text{CaCO}_3} = 100 \text{ kg/kmol}$, $M_{\text{MgCO}_3} = 84.3 \text{ kg/kmol}$, and $M_{\text{CO}_2} = 44 \text{ kg/kmol}$.

- (a) What is the fractional atom economy for the process if lime is the desired product? (Hint: You can choose any convenient basis for your calculation, such as, 100 lb_m of limestone.)
- (b) If limestone costs \$87/ton and lime can be sold for \$125/ton, what is the profit (or loss) rate (\$/day) for a lime kiln producing 250 ton/day of lime? Be sure to indicate whether it is profit or loss. (Hint: How much limestone is required to make 250 ton/day of lime?) 1 ton = 2000 lb_m
- (c) If for this process, it costs \$75/ton of carbon dioxide for the proper disposal of CO_2 waste, how does this change the result in part (b)?



We can see the molar ratio 1:1:1:1:2

Ratio between limestone and lime is:

$\frac{53.624}{100}$

Cost of making limestone = 987 $\frac{53.624}{100} =$